1 IoT and ML-based motor monitoring device

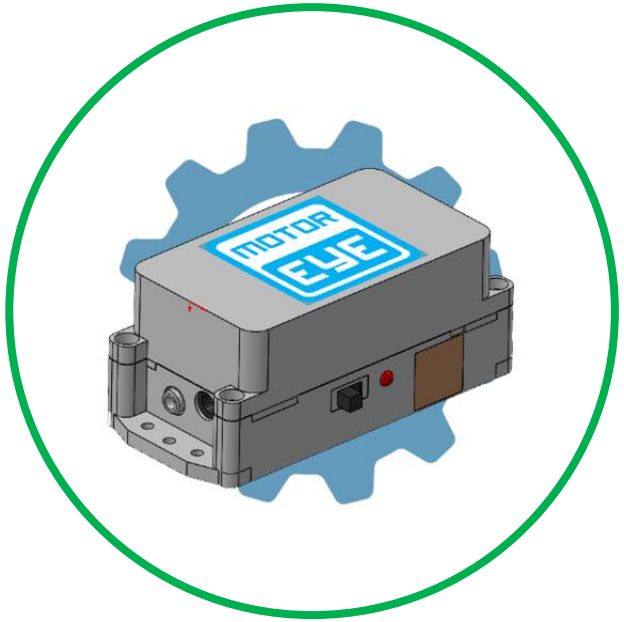


MOTOR EYE

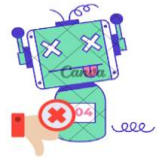


IoT and ML-based motor monitoring device

❑ Goal



- Measure parameters of the motor, engine
- Detect the abnormal signals
- Provide information for users on real-time
- Compact and easy to install



Detect unusual signs promptly



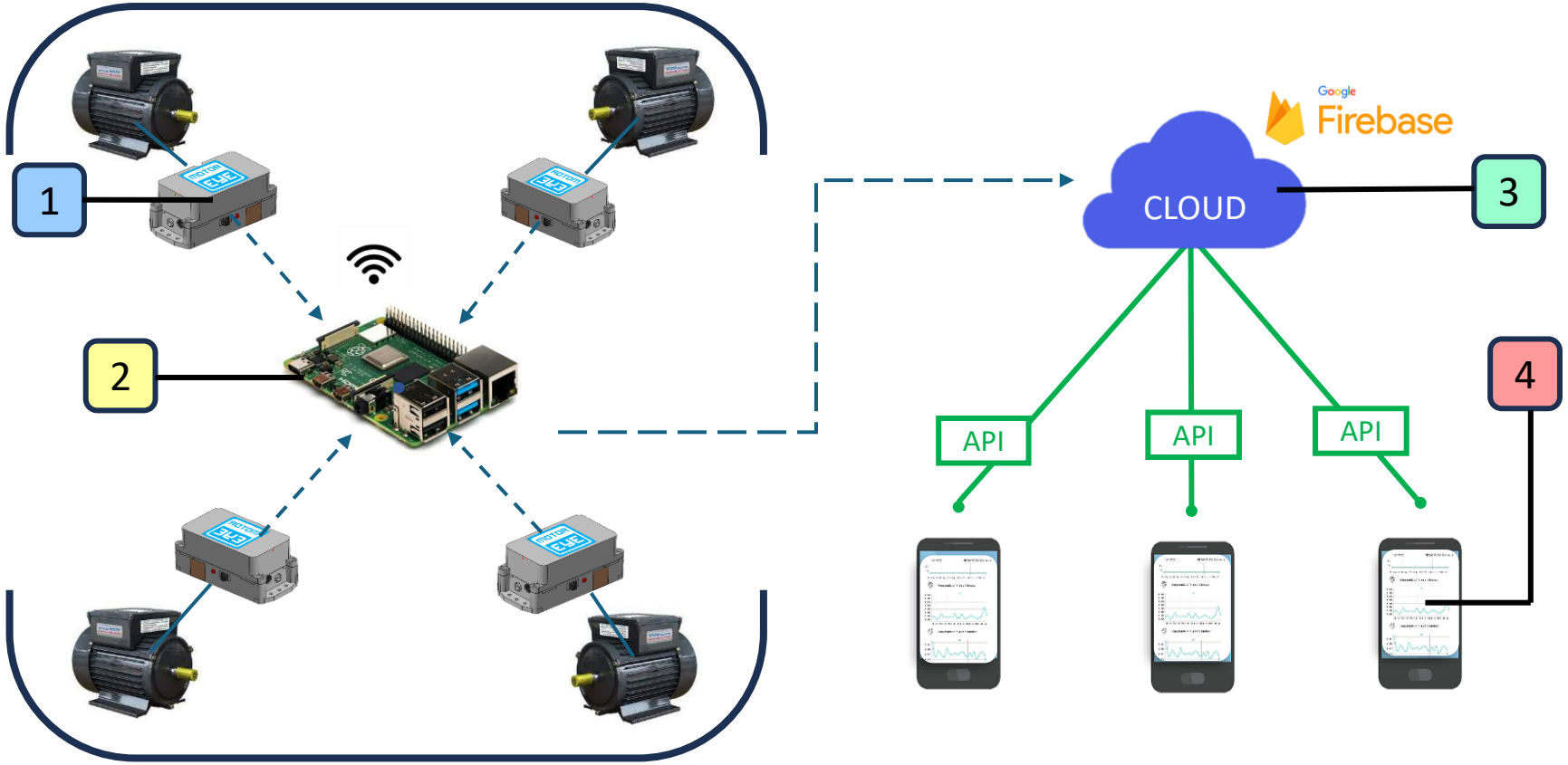
Ensure the machine's realiability



Increase life of machines

IoT and ML-based motor monitoring device

❑ Overview of the system



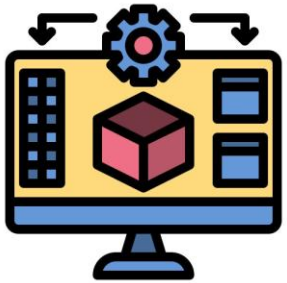
IoT and ML-based motor monitoring device

❑ My contribution



Team leader

- Deviding tasks and managing the project
- Requiring weekly reports



Hardware

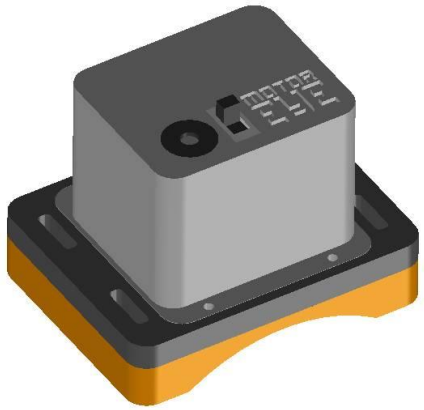
- PCB designing and manufacturing
- 3D shell designing and manufacturing

IoT and ML-based motor monitoring device

❑ Hardware designing and manufacturing



Version 1



Version 2

IoT and ML-based motor monitoring device

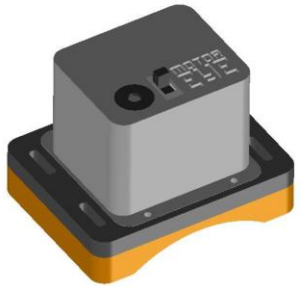
❑ Hardware designing and manufacturing

Version 1



- High accuracy
- Large battery capacity
- Large data storage
- Ensure safety

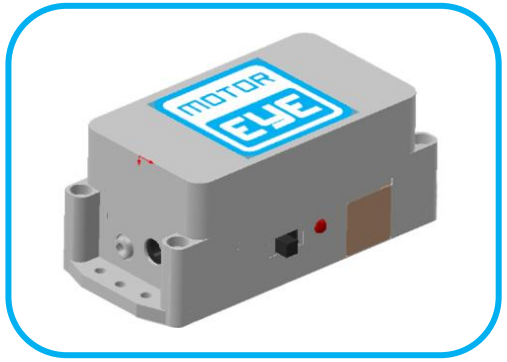
Version 2

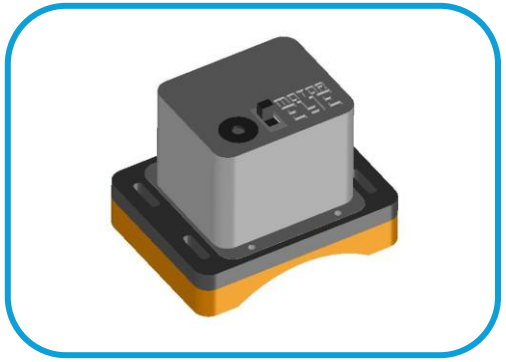


- Low cost
- Compact, minimum size
- Wide measuring range

IoT and ML-based motor monitoring device

❑ 2 version comparation

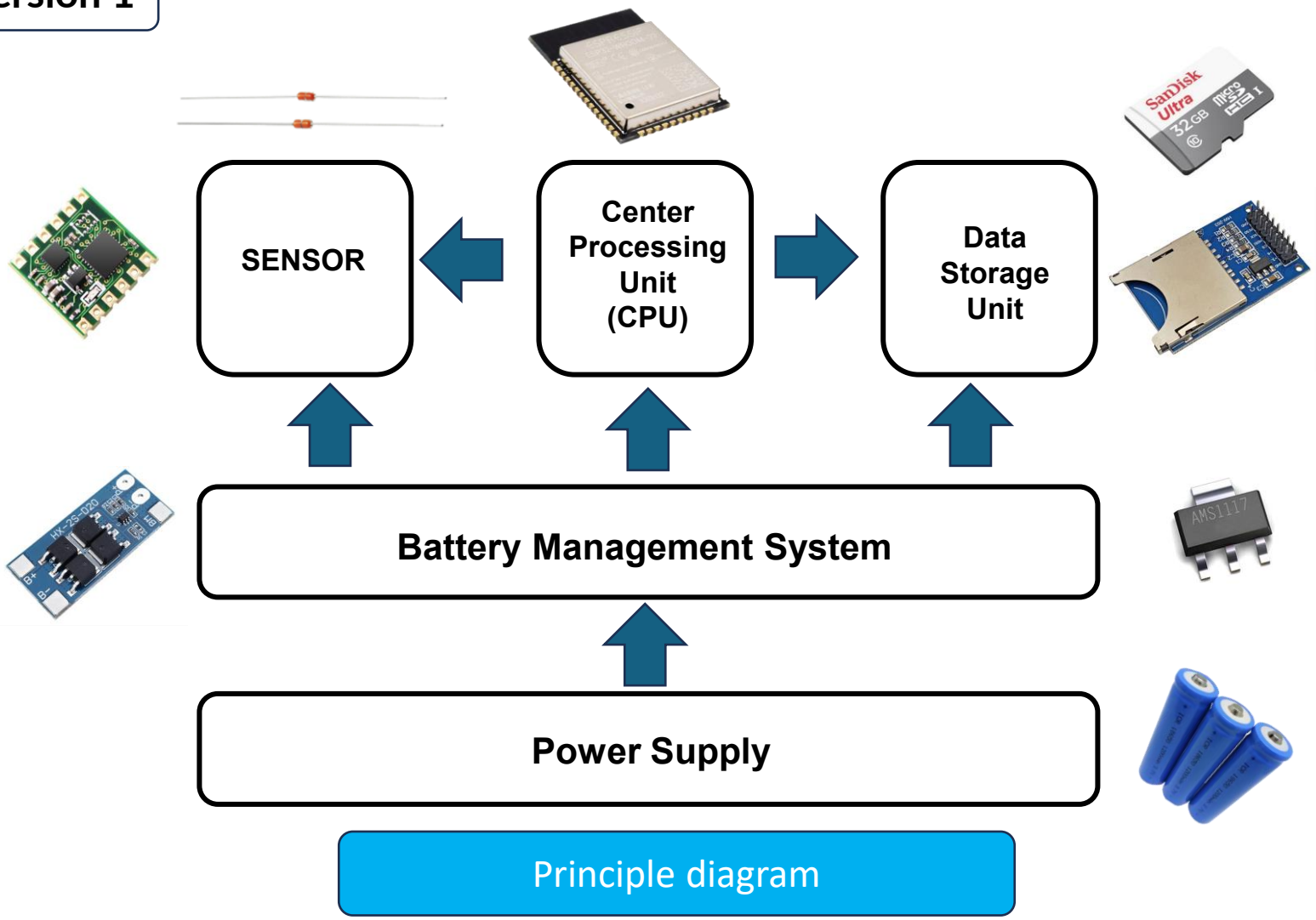




Version	Version 1	Version 2
Size	100 x 40.5 x 46 mm	70 x 55 x40.9 mm
Frequency range	0-150 kHz	0-400 kHz
SD card	Integrated	Not integrated
Battery management circuit	Integrated	Not integrated
Battery capacity	3600 mAh to 7200 mAh	1200 mAh
Recharging method	Replacing, charging	Replacing
Internal filter	Integrated	Not integrated
Estimated price	40$	15$

IoT and ML-based motor monitoring device

❑ Version 1



Principle diagram

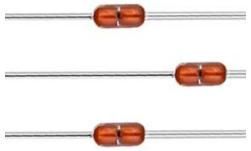
IoT and ML-based motor monitoring device

❑ Version 1

Size: 75x40 mm



IMU WT901



Thermistor



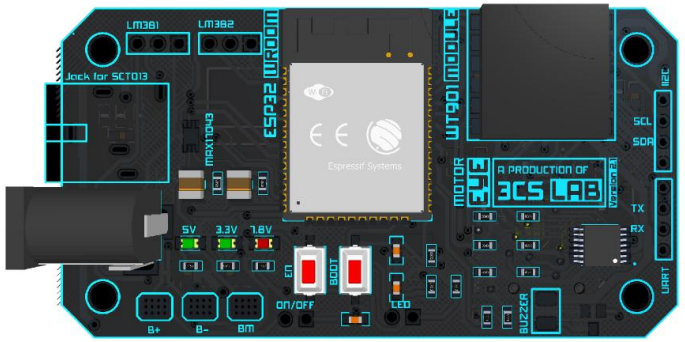
ESP32



AMS1117 – 5V

MOTOR EYE

Version 1



Frontside



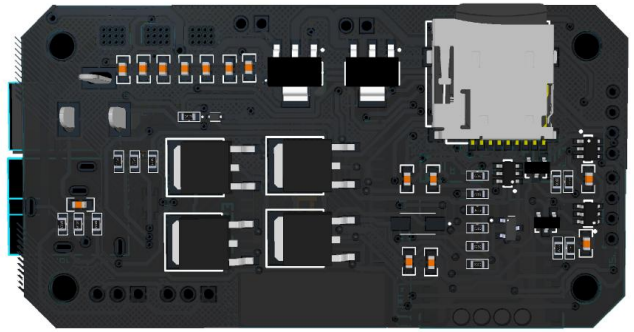
Battery manager system



18650 battery



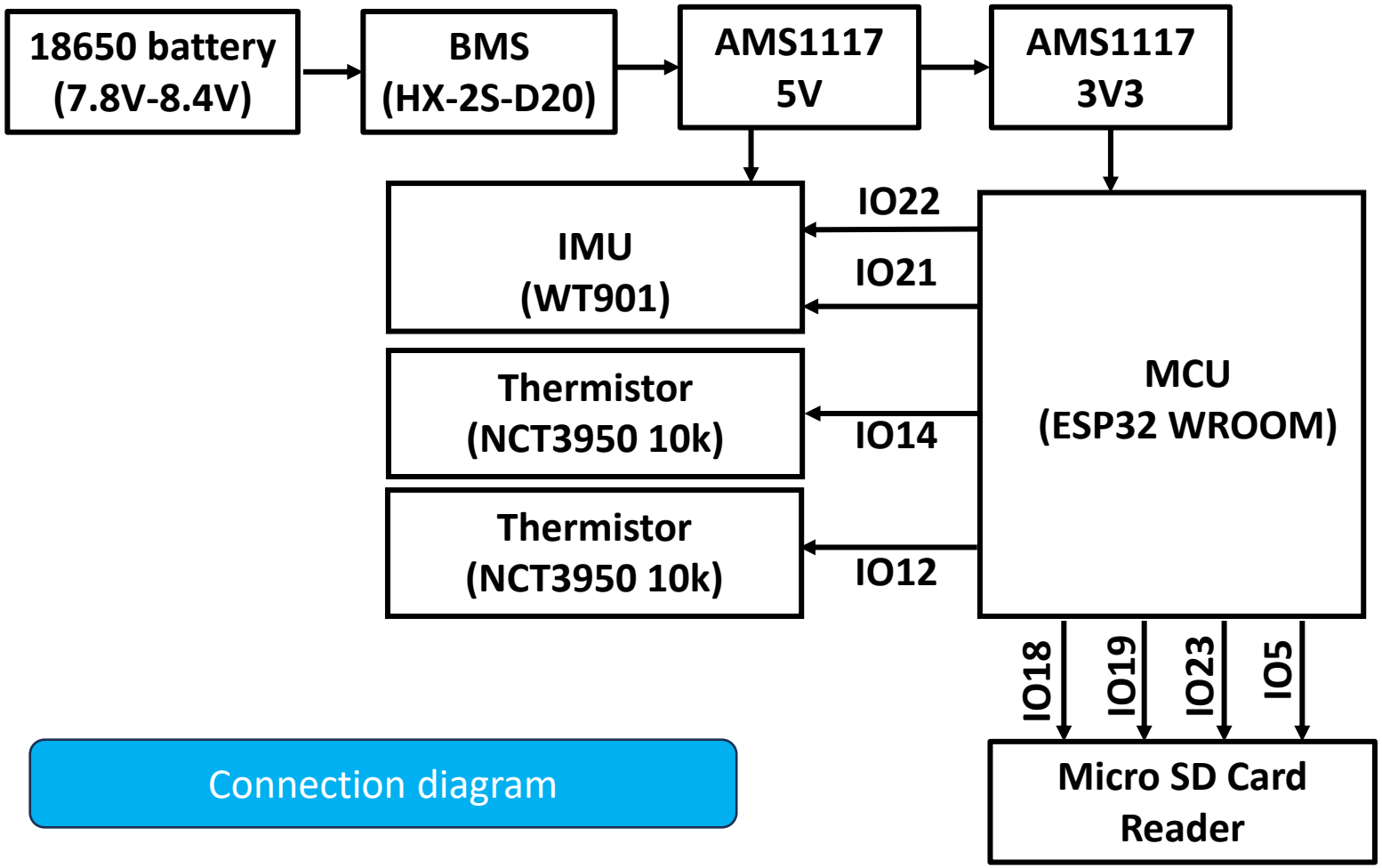
SD Card & Reader



Backside

IoT and ML-based motor monitoring device

❑ Version 1

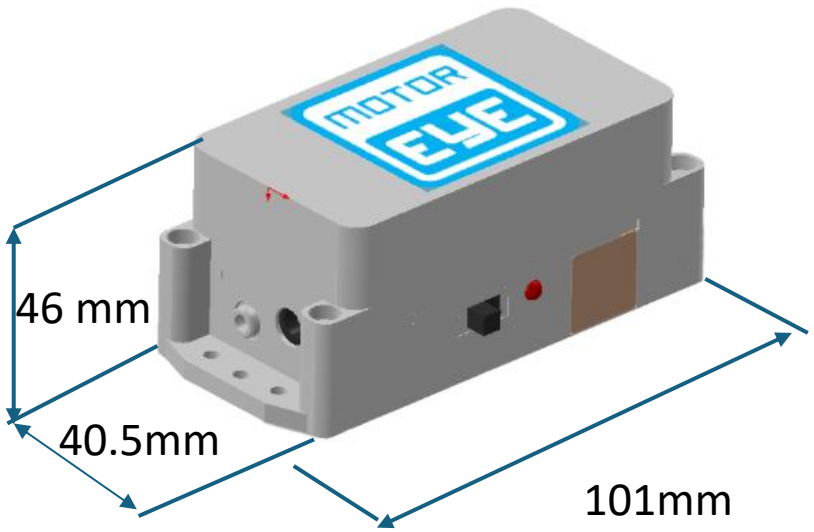


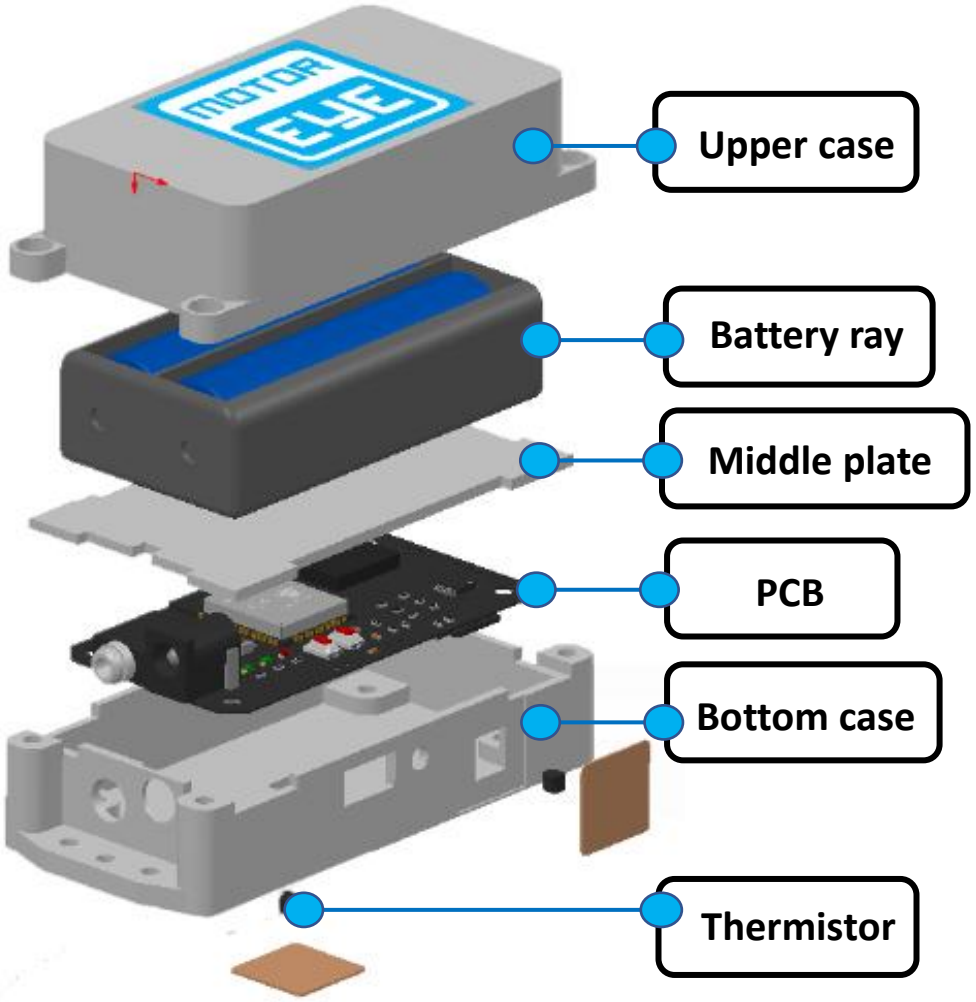
Connection diagram

IoT and ML-based motor monitoring device

❑ **Version 1**

3D model

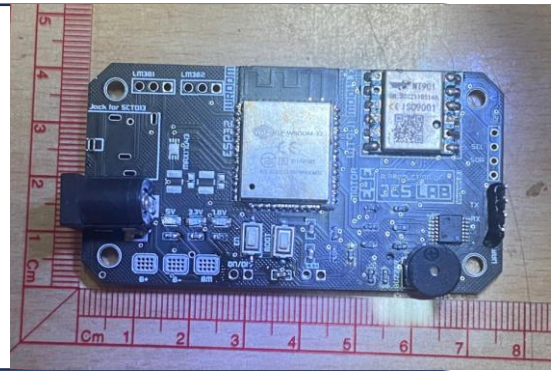




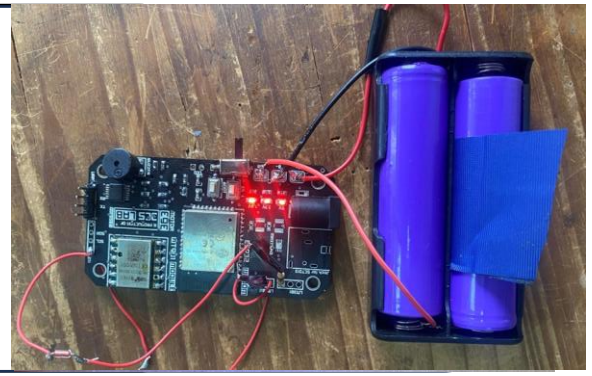
IoT and ML-based motor monitoring device

❑ Version 1

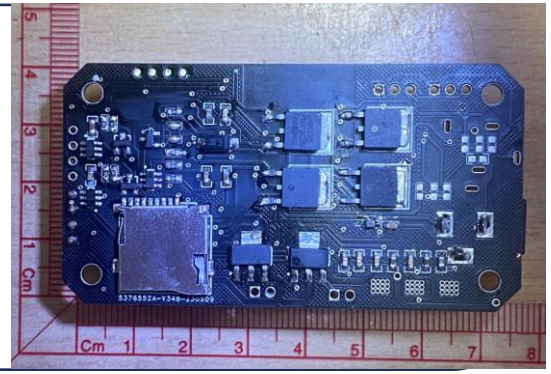
Front side of PCB



Completed PCB



Back side of PCB



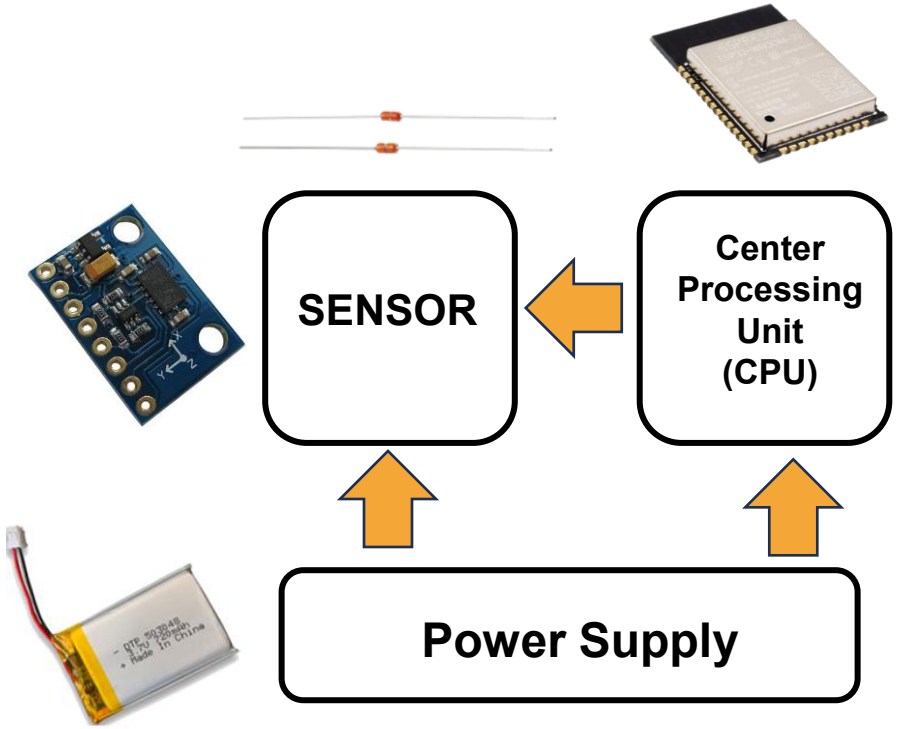
Completed device



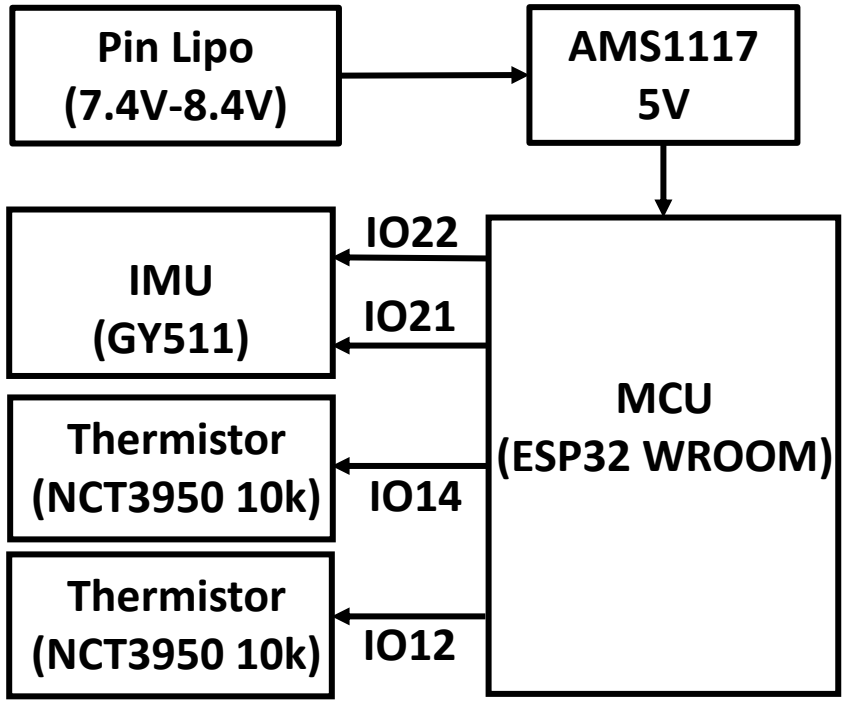
Fabricated product

IoT and ML-based motor monitoring device

❑ Version 2



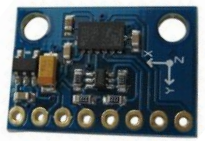
Principle diagram



Connection diagram

IoT and ML-based motor monitoring device

❑ **Version 2**



IMU GY511



ESP32



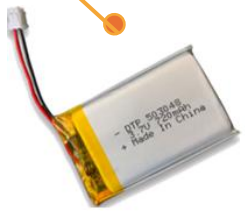
AMS1117 – 5V

MOTOR EYE

Version 2

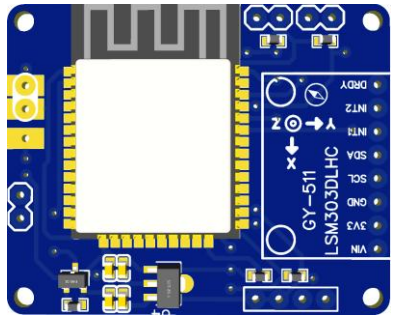


Thermistor

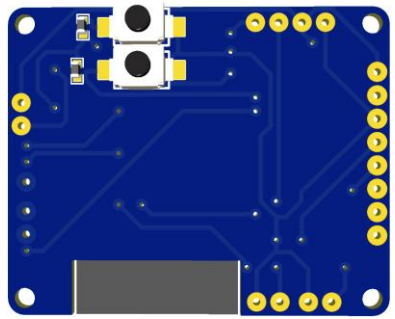


Li-po battery

Size: 45 x40 mm



Frontside

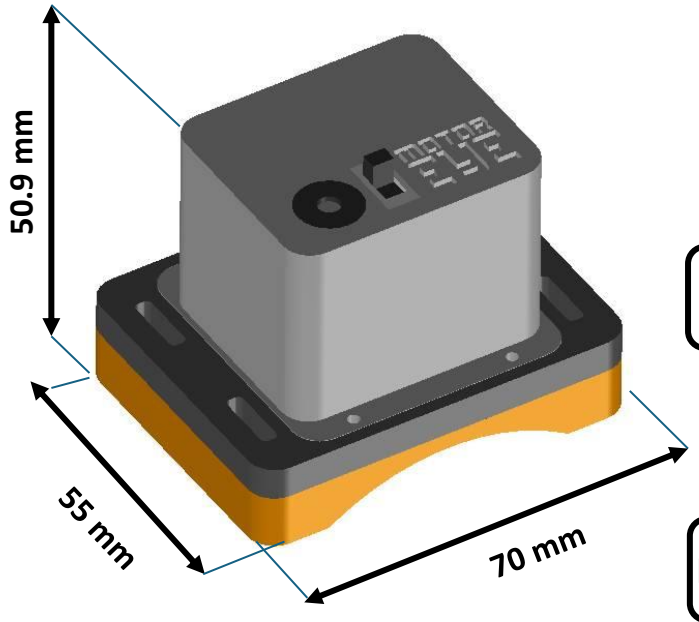


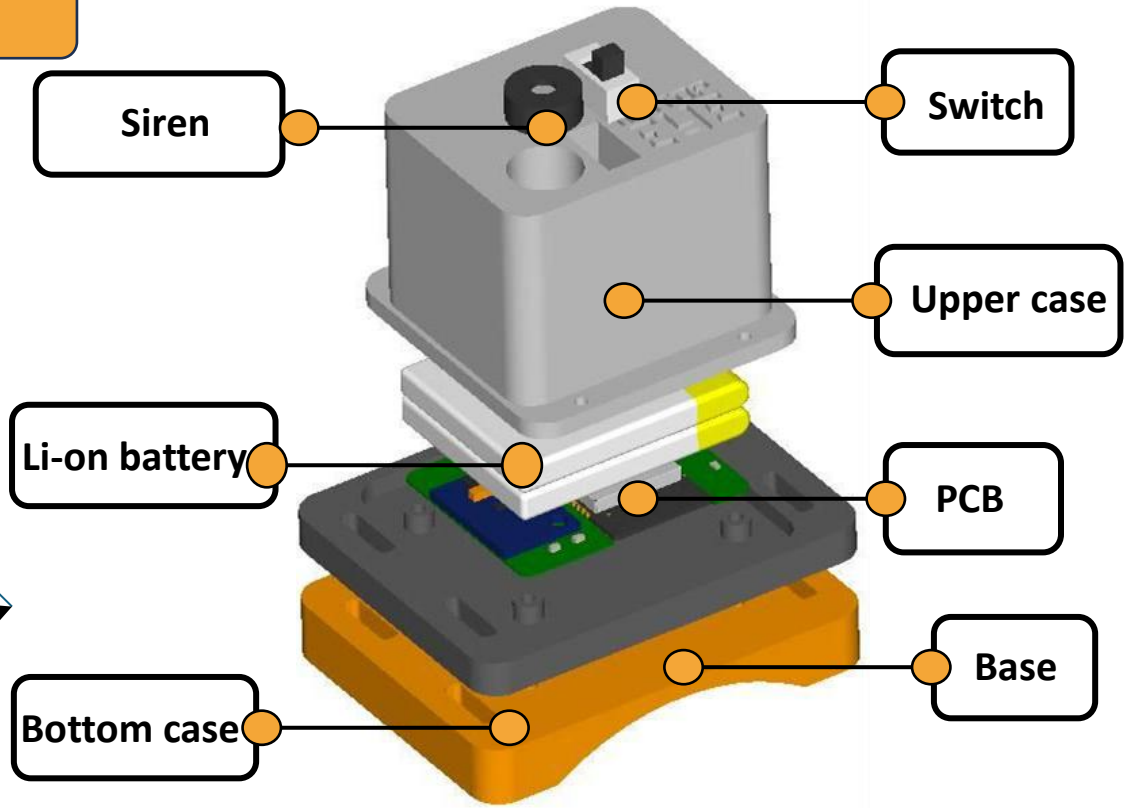
Backside

IoT and ML-based motor monitoring device

❑ Version 2

3D model

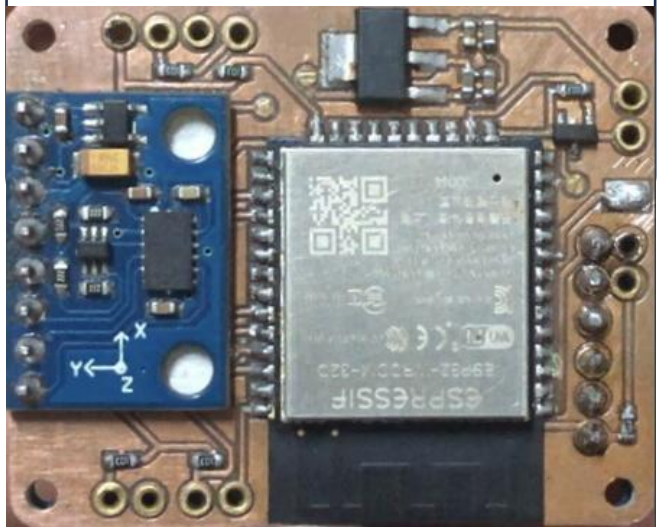




IoT and ML-based motor monitoring device

❑ Version 2

Completed PCB



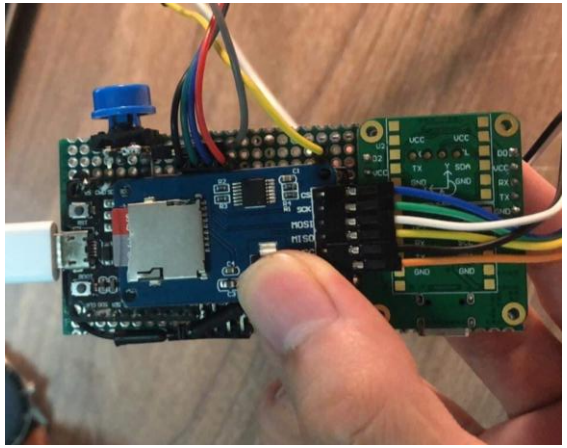
Completed device

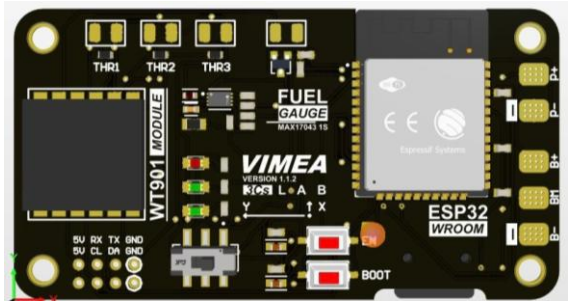


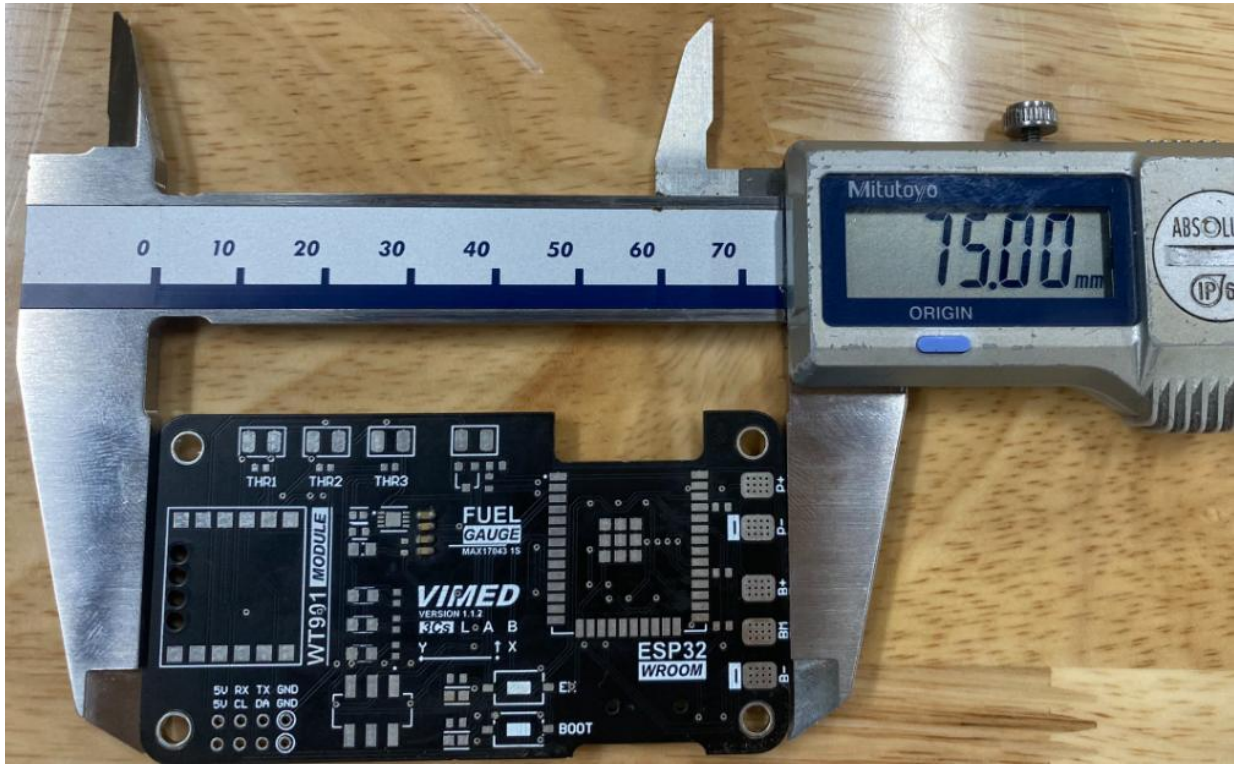
Fabricated product

IoT and ML-based motor monitoring device

❑ Other tested version





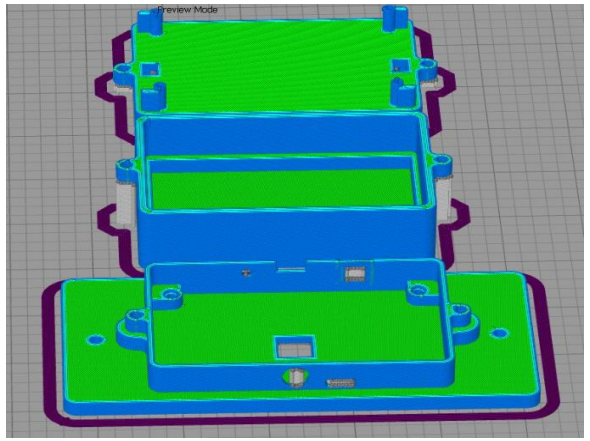


Other PCB versions

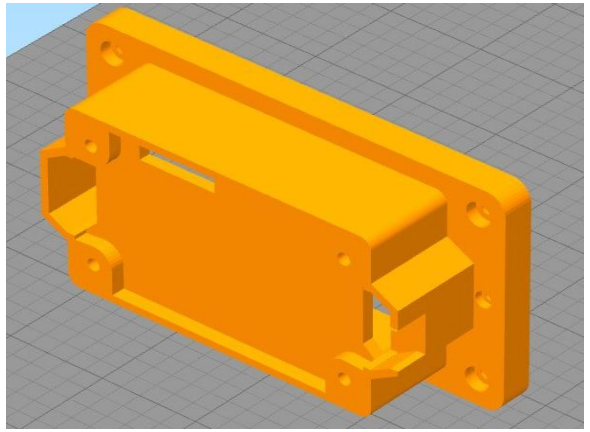
IoT and ML-based motor monitoring device

❑ Other tested version

Other 3D design & manufactured version









IoT and ML-based motor monitoring device

❑ Result



Product's efficiency:

- The manufactured product work well and stably
- High accuracy:
  - Over 97.5% with the ability to detect abnormalities.
  - Over 95% with the rotation speed measurement function





Achievements:

- A+ as my graduation thesis.
- Second prize in 2023 Scientific Research Conference organized for the 40th time by Hanoi University of Science and Technology.
- Interviewed and featured in the newspaper at the school's Sciencetific exhibition. (For more detail: LINK)



Publications:

- 1 international conference paper: "Development of IoT-based device for non-intrusive measurement of rotational speed of induction motors" Proceeding of the European Alliance for Innovation International Conference on Renewable Energy and Sustainable Manufacturing, 2023, pp. 01-11.
- Using to conduct another paper.